

Assignment 4: Derivations and ADTs

CAS CS 320: Concepts of Programming Languages

due Thursday, October 1 by 8:00PM

For the OCaml problems, you'll put your solutions in the file `lib/assign4.ml`. See the file `test/test_assign4.ml` for example behavior of each function. You'll submit your solutions via Gradescope, where your code will be checked against a collection of hidden tests. You must pass at least a third of the Gradescope unit tests on each problem for the assignment to be considered attempted. Your code must also follow the guidelines of the course for it to be considered attempted (i.e., it must use the course standard library, and it may not use any references, arrays, mutable fields, or for/while loops).

For the written problems, you'll submit your solutions via Gradescope as a single PDF file. They must be hand-written and exceptionally neat, and the final answer to each problem must be clear, e.g., surrounded by a box. You must cite your sources per the instructions in the course manual.

You're required to submit solutions to every problem in this assignment to receive full credit, but we will only provide detailed feedback on **bolded** problems. Since the assignment is graded on participation, we ask that you *try at least the bolded problems with as few external aids as possible*. You should think of these problems as simulated quizzes or exams. *You won't be penalized for mistakes*, so it's in your best interest to do these problems on your own; the detailed feedback we provide will generally be more useful (for you and for us) if you submit your own work.

OCaml Exercises

These problems are from the OCaml Exercises linked on the course webpage.

2.30 Split by Whitespace, Correctly

3.1 Walking Distance

3.4 Grouping Integers and Strings

3.(7-9) Binary Tree Insertion, Flattening Binary Trees, Sorting by Flattening

Written Problems

1. Using only anonymous functions and applications, determine an expression of the following type. In particular, you may not use type annotations, recursion, or assertions.

```
((('a -> 'b) -> 'b) -> 'c) -> 'a -> 'c
```

- (2) Consider the following inference system.

$$\frac{}{\text{foo} \diamond \text{bar}} \text{FOO BAR} \qquad \frac{}{\text{cat} \diamond \text{dog}} \text{CAT DOG} \qquad \frac{e_1 \diamond e_2}{e_2 \square e_1} \text{SWAP}$$
$$\frac{e_1 \diamond e_2 \quad e_3 \square e_4}{e_1 \ e_4 \square e_2 \ e_3} \text{COMBINE}$$

In this inference system an expression is a nonempty sequence of the words **foo**, **bar**, **cat**, and **dog**. There are two kinds of judgments: $e_1 \diamond e_2$ and $e_1 \square e_2$ where e_1 and e_2 are expressions. Give a derivation of the judgment **foo bar dog** $\square$ **bar foo cat**.¹

3. Determine a type τ and a minimal context Γ such that the following typing judgment is derivable in 320Caml. Also provide a derivation.

$$\Gamma \vdash \text{g (f (g 0))} : \tau$$

Is there more than one choice of τ and Γ ? Justify your answer.

- (4) Determine a type τ such that the following typing judgment is derivable in 320Caml. Also provide a derivation.

$$\{\} \vdash \text{let x = true in 3 + (if false || x then 2 else 1)} : \tau$$

¹Errata 2026-09-27 08:00PM: There was a typo in the conclusion of the (combine) rule.

Then determine a value v such that the following semantic judgment is derivable in 320Caml.

`let x = true in 3 + (if false || x then 2 else 1) ↓ v`

Also provide a derivation.

5. Give a derivation of the following typing judgment in 320Caml. *Note:* This will take some space.

`{f : int -> bool} ⊢ (if f 2 then (fun x -> true) else f) 3 : bool`